EXTENDING A RATIONAL MAP FROM A REGULAR SURFACE ALONG BLOWUPS IN POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to a finite complement by extension across codimension one, the proper modification over which the rational map is defined, and the domination by blowups in closed points.

Editorial note. The label lemma-extend-rational-map-blowing-up names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-extend-rational-map-blowing-up –start 1181 –end 1207 –local-id 28117 returned RESULT=OK (L28117 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Extending a rational map across points of codimension one. Morphisms, the lemma that a rational map is defined at all points of codimension one when the target is proper and separated. Reduces to the case where the complement of the domain is a finite set of closed points.
- Extending a rational map after a modification. Divisors, the lemma producing a proper modification over which the rational map is defined. Produces a proper morphism to the source which is an isomorphism over the domain of the rational map and carries a morphism agreeing with the rational map.
- Dominating a proper modification by blowups in closed points. Same chapter, the lemma dominating a proper morphism which is an isomorphism outside a finite set of regular surface points by a sequence of blowups in closed points above that set. Turns the modification into a sequence of blowups in closed points and provides the composed morphism to the target.

Source reference. Original lemma 0C4V: sources/stacks-chow/tags.txt, source lines 15240–15240. Prerequisite pointer invoked at proof line(s) 1201; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:resolve.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0AHI: sources/stacks-resolve/pinned-original-source.tex, source lines 1146–1146. Prerequisite pointer invoked at proof line(s) 1204; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BX7: sources/stacks-chow/tags.txt, source lines 14970–14970. Prerequisite pointer invoked at proof line(s) 1197; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a scheme over S which is regular and has dimension 2. Let Y be a proper scheme over S. Given an S-rational map $f : U \to Y$ from X to Y there exists a sequence*

$$X_n \to X_{n-1} \to \ldots \to X_1 \to X_0 = X$$

and an S-morphism $f_n : X_n \to Y$ such that $X_{i+1} \to X_i$ is the blowing up of X_i at a closed point not lying over U and f_n and f agree.

Proof. We may assume U contains every point of codimension 1, see Morphisms, Lemma 0BX7. Hence the complement $T \subset X$ of U is a finite set of closed points whose local rings are regular of dimension 2. Applying Divisors, Lemma 0C4V we find a proper morphism $p : X' \to X$ which is an isomorphism over U and a morphism $f' : X' \to Y$ agreeing with f over U. Apply Lemma 0AHI to the morphism $p : X' \to X$. The composition $X_n \to X' \to Y$ is the desired morphism. □